Statistical Report

Otto

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Statistical Report on Fertility Data in Singapore

Abstract

Introduction

Singapore has long experienced a long-term decline in fertility of its population. This has many knock on affects to the economic and social structures of the country. Decreasing fertility can raise concerns about an ageing population, lack of labour supply and demographic sustainability. Therefore forecasting fertility is important for the governments future understanding of policy and planning.

This report focuses on two annual time series, Total Fertility Rate and Total Live Births. These series will be henceforth refered to as TFR and TLB respectively. TFR is the number of children an average woman would have in her lifetime if she experienced current age specific fertility rates. TLB is the total number of live births experienced in Singapore over the time period, in our case a year. TFR reflects overall fertility more directly, whereas TLB is affected by population size, marital rates, immigration and other demographic factors.

This leads us to our research question of **Can time series models trained on data from 1960–2012 accurately forecast Singapore's continuing fertility decline over 2013–2024?**

ARIMA and SARIMA models are used as they can

TO INLCUDE WHEN INTRODUCTION IS DONE

Inlude nessacary abbreviations when writing it up

4. Justify SARIMA period 12 in Introduction (page 3) The periodogram doesn't obviously point to period 12, so you need literature to justify it. Add a paragraph in the Introduction discussing the Chinese zodiac Dragon year effect on Singapore births. This is what makes the SARIMA(p,d,q)(P,D,Q)[12] models substantively motivated rather than just data-driven.

Methods

The modelling workflow was designed to answer our forecasting needs. The TFR and TLB series were extracted from the dataset, reshaped into annual time series from 1960–2024 and split into a training period from 1960–2012 and a testing period from 2013–2024.

Time plots and the Kwiatkowski-Phillips-Schmidt-Shin (KPSS) tests of the two series were used to check for trends, changes in variance and the transformations required before modelling. Both series showed evidence of non-stationarity and since ARIMA models require the time series to be stationary, transformation of the series was necessary prior to modelling.

The log transformation was used to stabilise variance and make the model proportional rather than absolute changes. Differencing was used to remove the trends shown inside the data.

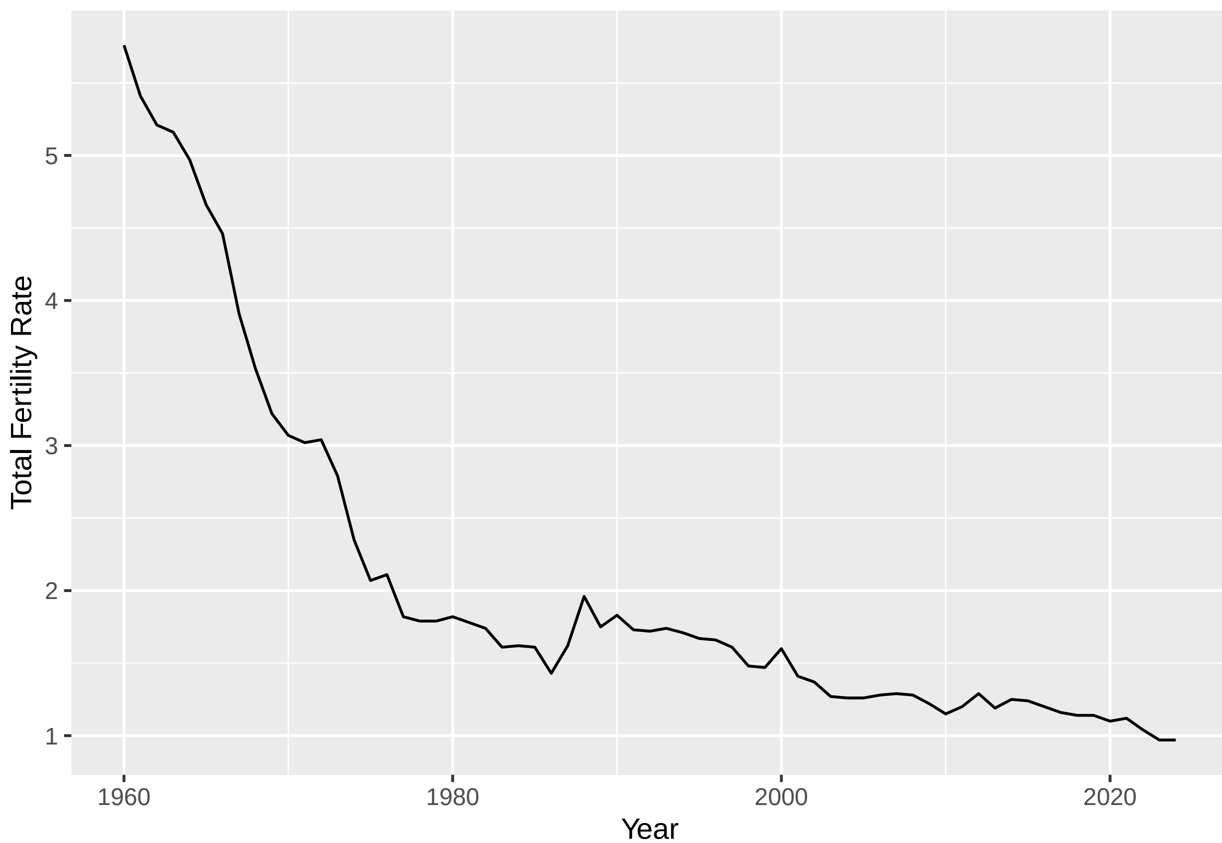


Figure 1: Annual Total Fertility Rate in Singapore

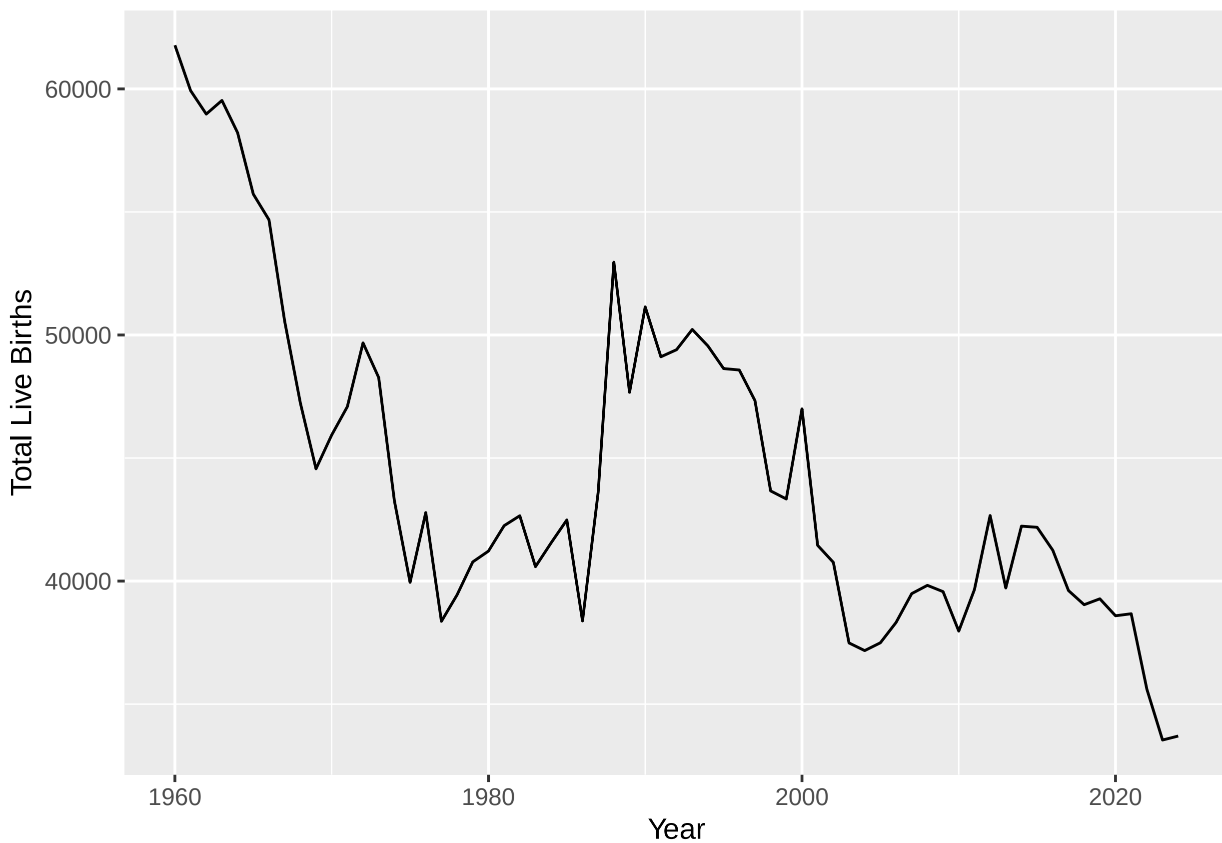


Figure 2: Total Live Births in Singapore

Both series were then transformed using either just the log or the log with first or second differencing. These transformed series were then tested by the KPSS test to ensure an improvement of the stationarity of the series.

Two transformations showed potential as being stationary (first-differenced and logged, and second-differenced logged). With the KPSS test a small p-value shows evidence against stationarity. For TFR, the KPSS test gave that $p = 0.0406$ for the first-differnced logged series and $p = 0.1$ for the second-differenced logged series. For the logged TLB with both first and second differencing gave the same p-values of $p = 0.1$. So for TLB stationarity was not rejected for either transformation.

Despite the first difference of the TFR data having a KPSS test result below the significance threshold, we would prefer there to be less differencing as we do not want to over-difference the data and introduce noise in a relatively short data-set. We kept the second-differenced models as comparison as well so we could see another set of models in the results.

Because the transformed series and baseline residual diagnostics showed dependence around lags 11–13, higher-lag ARIMA models were also considered. These models were included to capture longer-lag autoregressive dependence that was not represented by the low-order baseline models.

SARIMA-style models with period 12 were also considered. Although the data are annual, the period-12 structure was motivated by both the lag-12 dependence observed in the transformed series and the literature discussed in the Introduction on Chinese zodiac-related birth timing effects in Singapore. These models were included to assess whether a 12-year pattern could be represented more simply than by unrestricted high-order ARIMA models. For the SARIMA candidates, the ACF and PACF plots of the transformed series were inspected to guide the initial period-12 model specification.

Models were only considered viable if their residuals were approximately white noise. This again was assessed with ACF and PACF, portmanteau tests and residual plots. Among viable models, the final model selection was based on AIC and forecast accuracy over the 2013-2024 period. Forecast accuracy was assessed with RMSE and MAE, with formulas provided in the appendix.

Results

The ACF and PACF plots of the log-first-differenced series are shown in Figure 3. For both TFR and TLB the plots show a remaining dependence on lags 11-13, particularly at lag 12. This suggests that low order ARIMA models may not fully account for the higher order dependencies shown by Figure 3.

To assess the period-12 SARIMA candidates, the ACF and PACF of $(1 - B)\log(X_t)$ (Figure 3) and $(1 - B^{12})(1 - B)\log(X_t)$ (Figure 4) were used to guide the structure of our models. For both series, Figure 3 showed PACF spikes at lag-12 this suggested a seasonal AR term could be useful, while the lower lag ACF structure supported testing simple MA terms. This informed our decision for our first SARIMA candidate model to be SARIMA(0,1,1)(1,0,0)[12]. Nearby candidates were also fitted by varying the non-seasonal AR and MA orders and by comparing seasonal AR and MA terms.

In our seasonally differenced plot (Figure 4), lag-12 dependence remained even after the period 12 differencing. This indicated that differencing alone could not remove the period 12 dependence structure and supported the use of seasonal ARIMA models.

Firstly, we fitted some low-order ARIMA models without seasonal terms, however, as expected based on the lag-12 dependence identified with the ACF/PACF plots these models failed to produce white noise residuals. We next decided to fit some higher-order ARIMA models with seasonal terms as well as some SARIMA models with lag 12.

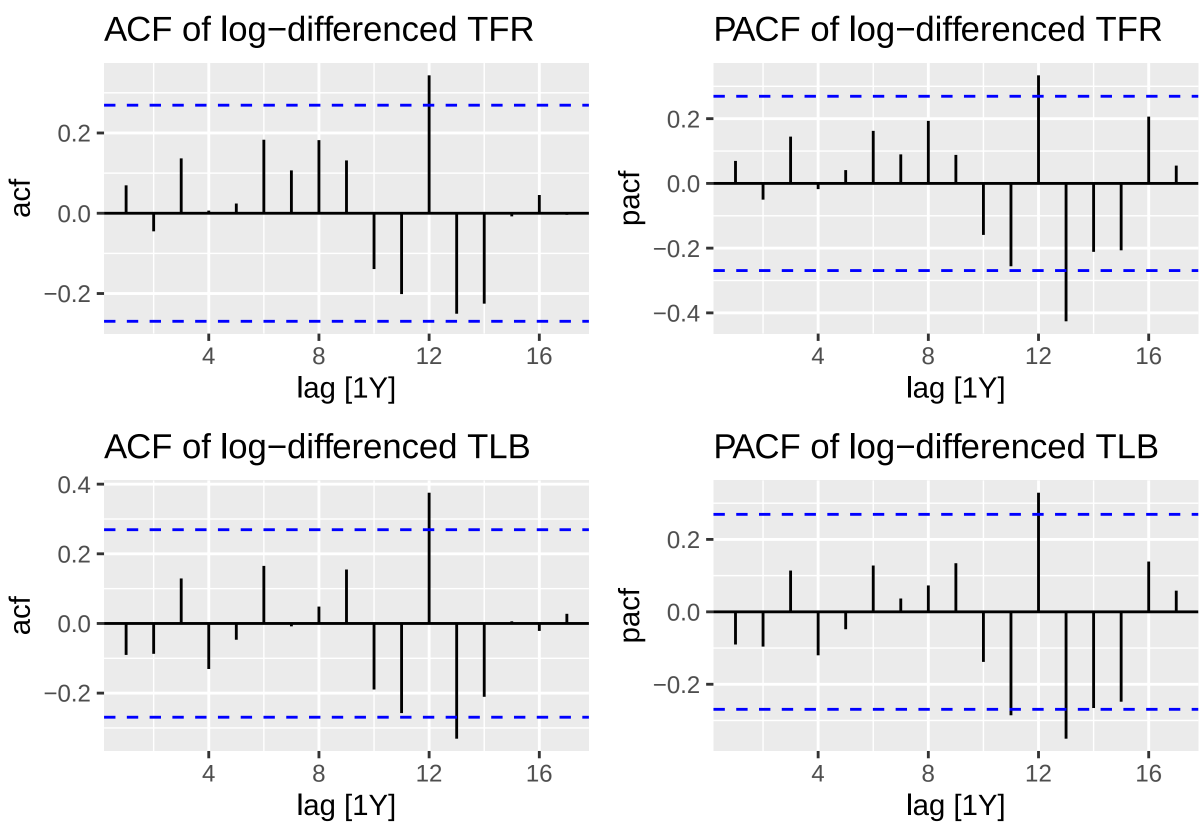


Figure 3: ACF and PACF plots of the log-first-differenced TFR and TLB series

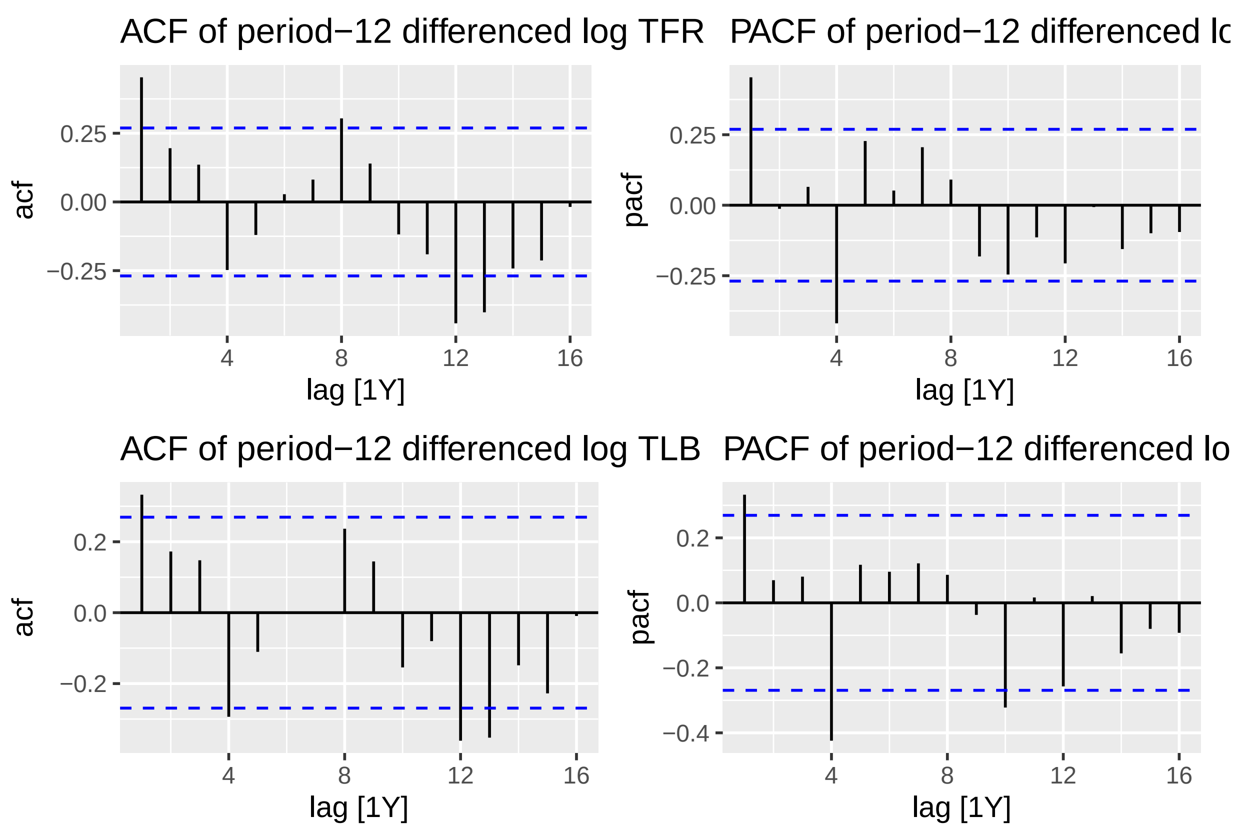


Figure 4: ACF and PACF plots of the log-first-differenced and period-12 differenced TFR and TLB series

Table 1: Ljung–Box residual tests for TFR and TLB candidate models

model_type	tfr_lb_pvalue	tfr_viable	tlb_lb_pvalue	tlb_viable
ARIMA(0,2,1)	0.005	No	0.003	No
ARIMA(1,2,1)	0.003	No	0.004	No
ARIMA(11,1,0)	0.000	No	0.001	No
ARIMA(12,1,0)	0.002	No	0.012	No
ARIMA(13,1,0)	0.020	No	0.045	No
ARIMA(13,1,1)	0.022	No	0.109	Yes
SARIMA(0,1,1)(0,0,1)[12]	0.027	No	0.028	No
SARIMA(0,1,1)(1,0,0)[12]	0.099	Yes	0.133	Yes
SARIMA(0,2,1)(1,0,0)[12]	0.004	No	0.049	No
SARIMA(1,1,1)(0,0,1)[12]	0.004	No	0.019	No
SARIMA(1,1,1)(1,0,0)[12]	0.108	Yes	0.115	Yes
SARIMA(1,2,1)(1,0,0)[12]	0.082	Yes	0.121	Yes

Table 1 shows all of our different models and their residual autocorrelation tests pvalues. The results how that most high order ARIMA models with lags 11-13 failed the Ljung-Box Portmanteau test for both series with the only exception being the ARIMA(13,1,1) for TLB. This suggests that adding higher order AR terms was not enough to remove residual autocorrelation. The SARIMA models with a period 12 seasonal AR term performed much better with SARIMA(0,1,1)(1,0,0)[12], SARIMA(1,1,1)(1,0,0)[12] and SARIMA(1,2,1)(1,0,0)[12] being viable models in terms of their residuals for both series.

Table 2: AIC, AICc and BIC values for TFR candidate models

.model	AIC	AICc	BIC
tfr_sar_011_100_log	-137.096	-136.596	-131.243
tfr_sar_111_100_log	-136.479	-135.628	-128.674
tfr_sar_121_100_log	-132.258	-131.388	-124.531

Table 3: AIC, AICc and BIC values for TLB candidate models

.model	AIC	AICc	BIC
tlb_sar_011_100_log	-146.736	-146.236	-140.882
tlb_sar_111_100_log	-145.079	-144.228	-137.274
tlb_sar_121_100_log	-138.646	-137.776	-130.918
tlb_ar13ma1_log	-141.446	-128.113	-112.178

Tables 2 and 3 show the AIC, AICc and BIC values for the viable TFR and TLB candidate models. For both series, SARIMA(0,1,1)(1,0,0)[12] had the lowest information criteria values among the viable candidates. However, these criteria should be interpreted with caution as the candidate sets both include models with different orders of differencing and as such cannot be directly compared across $d = 1$ or $d = 2$ models. With the research question in mind we are also preferring forecast accuracy for the 2013-2024 testing period as these in sample model fit results are not based on true out of sample forecast accuracy.

Table 4: Forecast accuracy for viable TFR models over 2013–2024

.model	RMSE	MAE	MAPE
tfr_sar_121_100_log	0.063	0.049	4.478
tfr_sar_011_100_log	0.098	0.073	6.945
tfr_sar_111_100_log	0.132	0.102	9.671

Tables 4 shows the forecast accuracy for the viable TFR models over 2013–2024 with 3 different forecasting metrics Root Mean Squared Error (RMSE), Mean Absolute Error (MAE) and Mean Absolute Percentage Error (MAPE). In the results we can see that SARIMA(1,2,1)(1,0,0)[12] has the lowest for each of the testing criteria therefore being the best prediction model for TFR. This leads us to select it as our preferred TFR forecasting model.

Table 5: Forecast accuracy for viable TLB models over 2013–2024

.model	RMSE	MAE	MAPE
tlb_sar_121_100_log	3035.560	2280.238	6.151
tlb_sar_011_100_log	3667.483	2752.134	7.542
tlb_sar_111_100_log	3914.953	2943.136	8.091
tlb_ar13ma1_log	10016.593	8587.467	23.304

Table 5 shows the forecast accuracy for the viable TLB models over 2013–2024. Like in the TFR the trained model that performs best over the testing dataset is SARIMA(1,2,1)(1,0,0)[12]. It has the smallest of our 3 criterion and as such will be our final TLB prediction model as well.

Figure 5 shows a plot of the chosen model for the TFR series, SARIMA(1,2,1)(1,0,0)[12], vs the actual observed data from 2000 to 2024. The black line is the observed data and the blue line is our predicted value for each year. The shaded blue area is our 80% prediction interval and shows the model forecast along with any uncertainity from the model. This figure supports the forecast accuracy results from Table 4 As we can see that our model correctly forecasts the continual fall of the fertility rate during the testing period. However, the model does not perfectly match each year to year movement, and should be interpreted as being able to capture the general trend of the data downwards.

Figure 6 compares our selected model, SARIMA(1,2,1)(1,0,0)[12], with the observed data from 2000 to 2024. The model forecasts a stabilisation in the TLB series, however, the observed series falls between 2021 and 2024. This shows that TLB is harder to forecast than TFR due to the chance that there may be more demographic variables which affect TLB that cannot be captured by a univariate time series model.

Overall, Figures 5 and 6 supports our final model choice of SARIMA(1,2,1)(1,0,0)[12] for both series. The model passed the residual autocorrelation test and had the best testing sample forecast accuracy values out of our viable models.

Discussion

The main finding of this analysis is that a SARIMA(1,2,1)(1,0,0)[12] is the preferred forecasting model for both TFR and TLB. The selection of a model with second differencing suggests that the decline of Singaporean birth rate and fertility is not just linear. Instead the decline appears to have steepened. This is evident in Figures 1 and 2, where the decline is especially pronounced in recent years. The $d = 2$ specification was able to react to the steepening decline better than the $d = 1$ was. Models which use $d = 1$ are generally more conservative and do not react as steeply. This could be an explanation for why our second-differenced model performed better for this short-term forecasting problem but we have to keep in mind that the model may be not as realistic over longer horizons. This is the general trade-off of using second-differencing: improved short-term responsiveness with the cost of stability in the long-term.

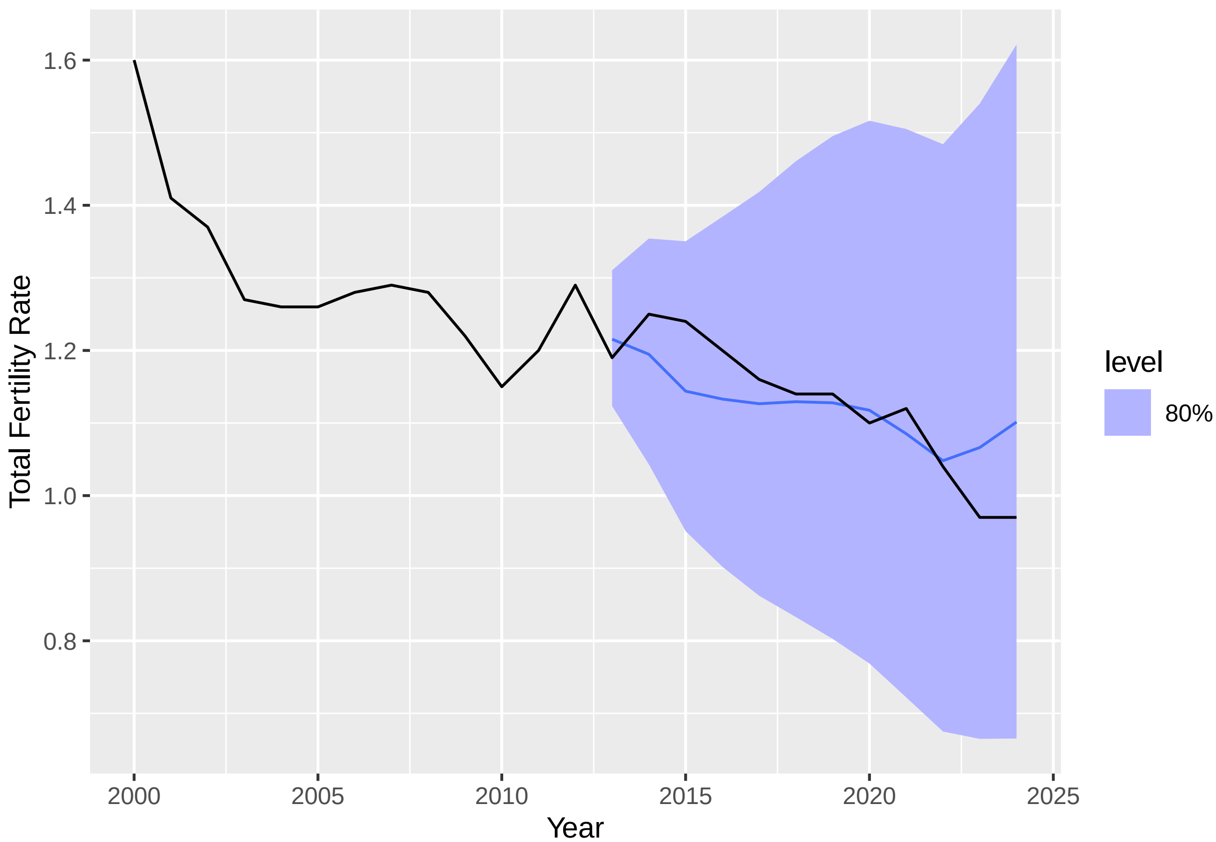


Figure 5: Selected TFR model forecast vs observed (2000–2024)

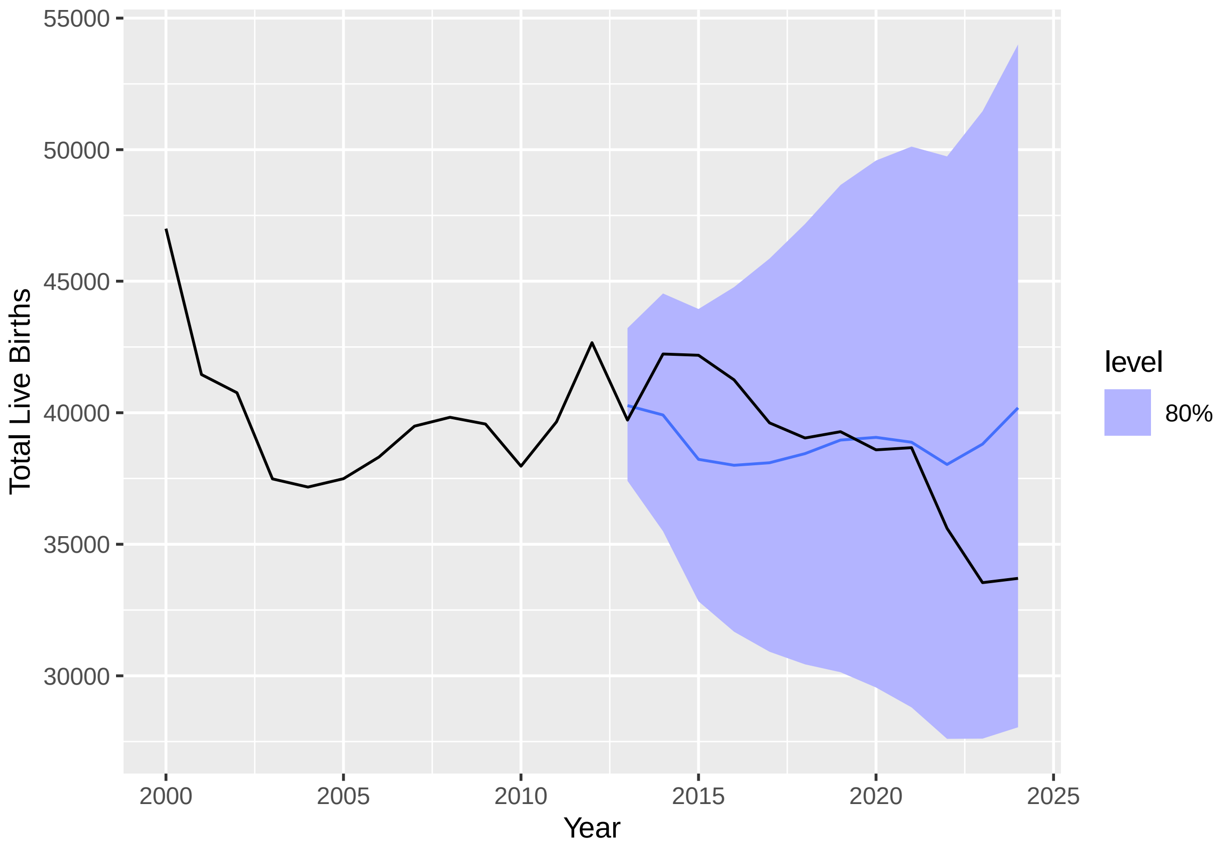


Figure 6: Selected TLB model forecast vs observed (2000–2024)

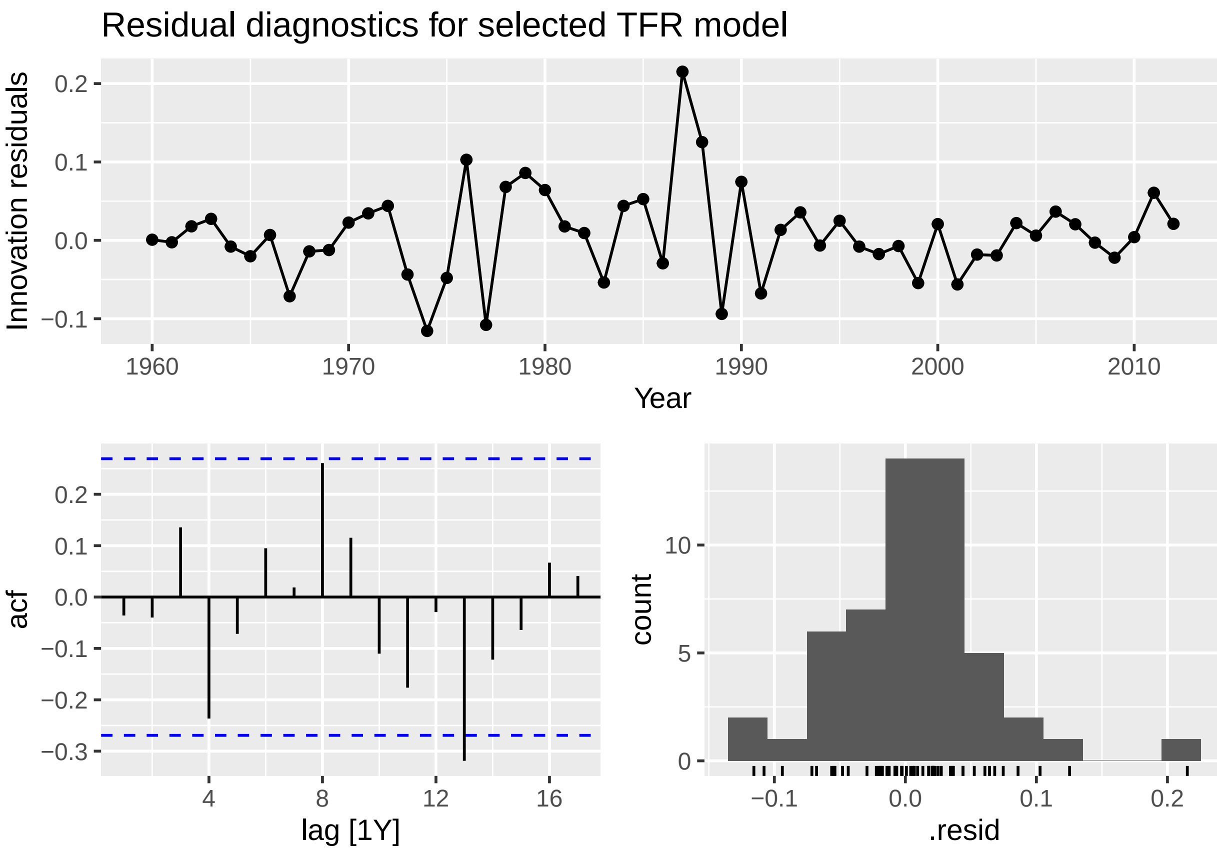
The model forecasts TFR more accurately than TLB. TLB being all the total live births in Singapore means that it is affected by total population and overall fluctuations in other demographic factors than just the birth rate. TFR is a more representative of the proportion of the population and is a rate which means that it follows a smoother demographic trend rather than just population size. Therefore, a good model predicting overall fertility rates can still deviate when predicting total births as there are more variables.

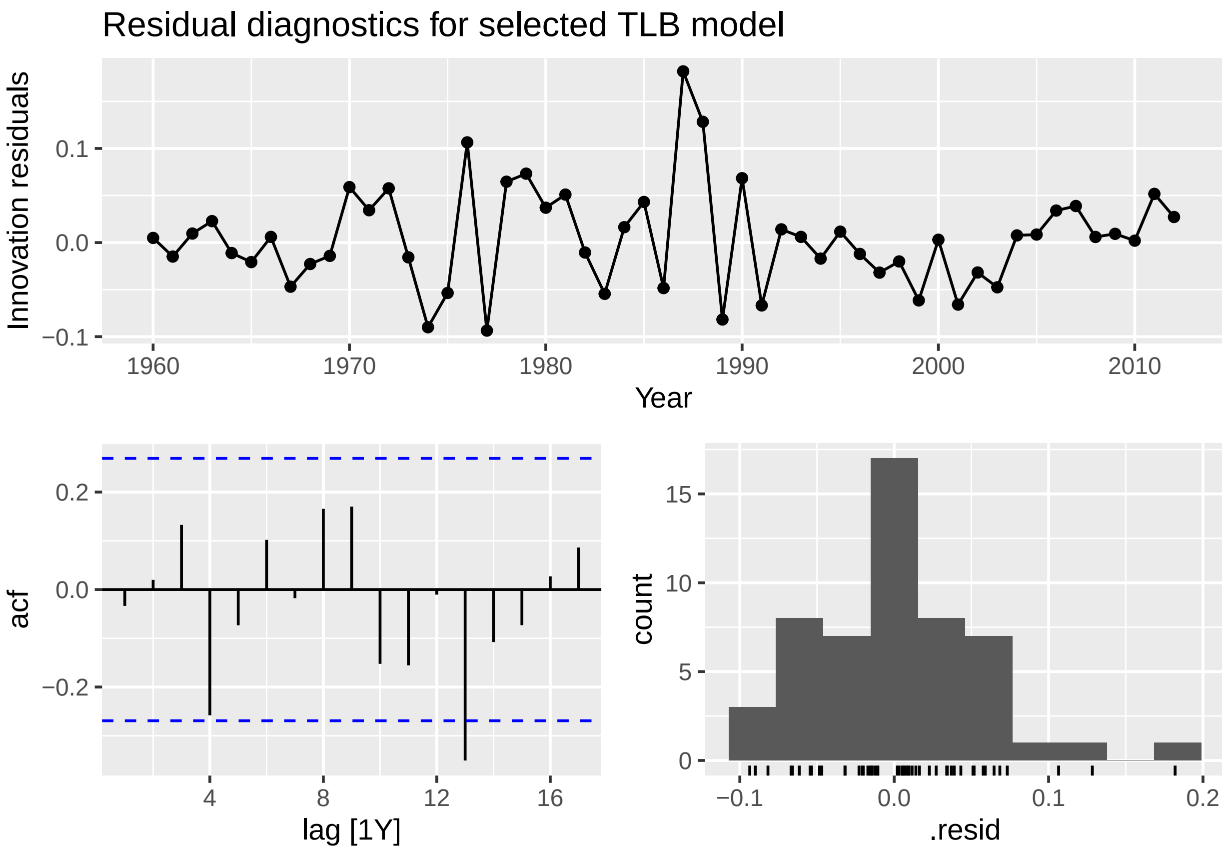
We see this the sharp decline can be attributed to the overall shock from COVID-19 and the affect on immigration into Singapore

stop of immigration during covid effects TLB not TFR. ## Conclusion

References

Appendix




Residual diagnostics for selected TLB model
Innovation residuals
Year
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